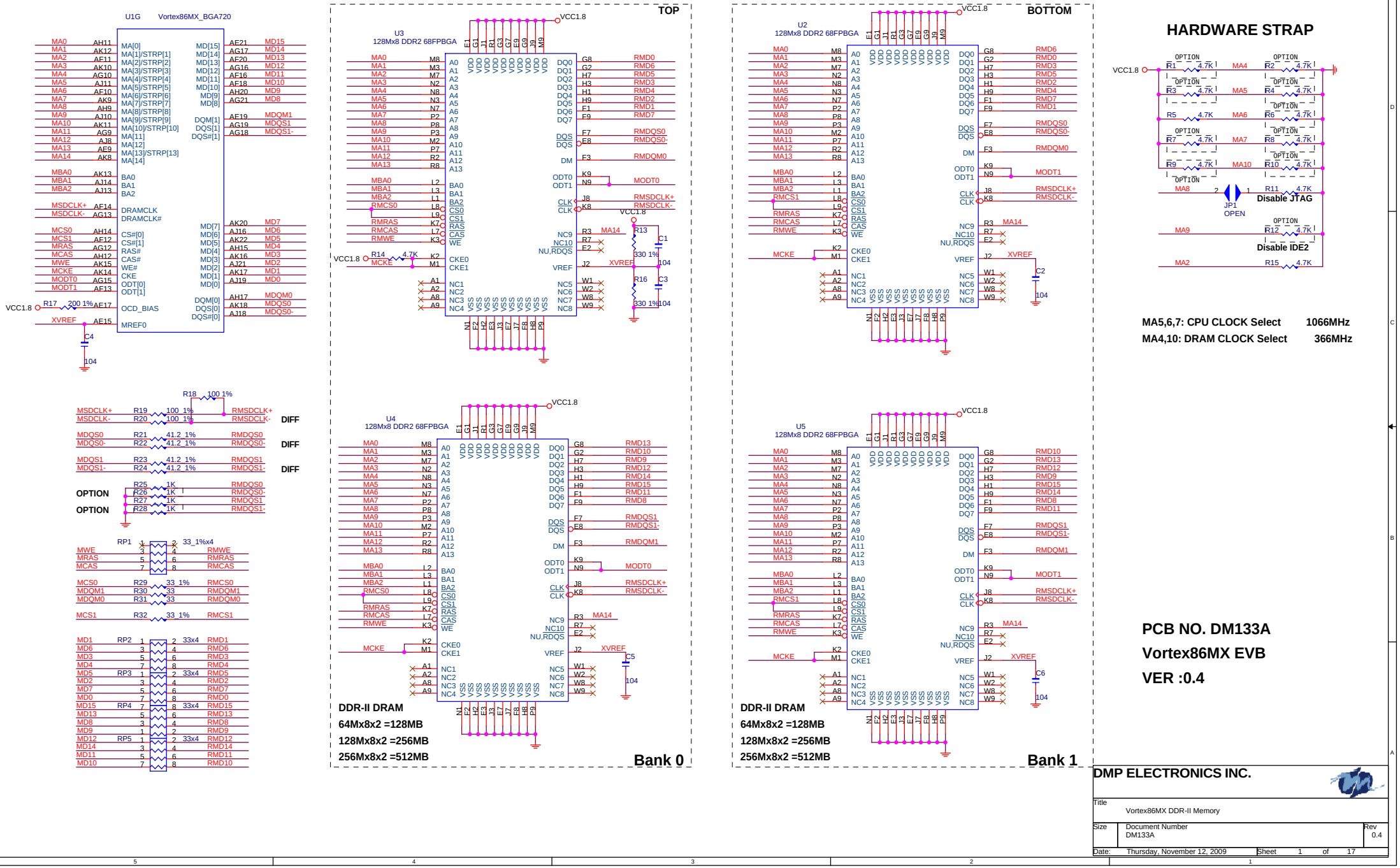
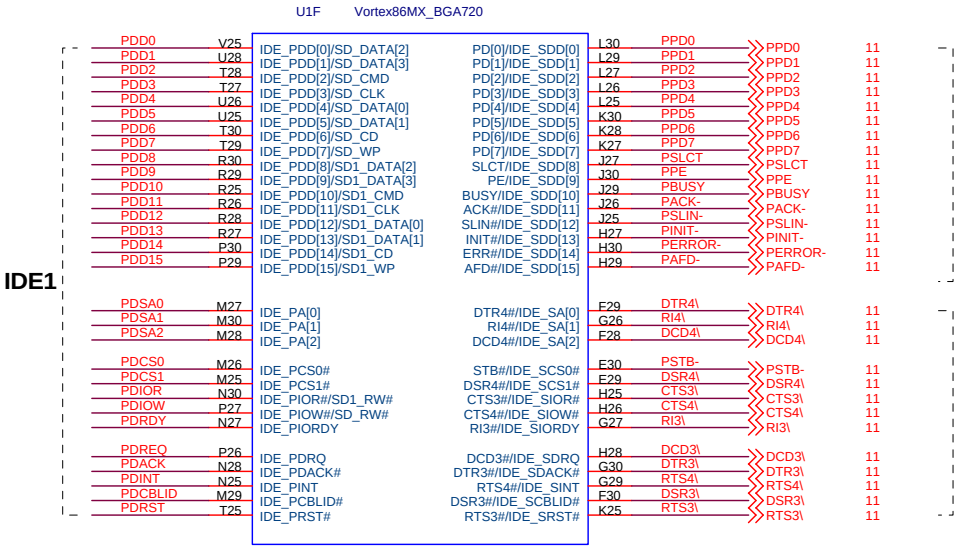

U1G Vortex86MX_BGA720
TOP
U3 128Mx8 DDR2 68FPBGA
U2 128Mx8 DDR2 68FPBGA
BOTTOM
U4 128Mx8 DDR2 68FPBGA
U5 128Mx8 DDR2 68FPBGA
VCC1.8
DDR-II DRAM
64Mx8x2 =128MB
128Mx8x2 =256MB
256Mx8x2 =512MB
Bank 0
Bank 1
DIFF
OPTION
HARDWARE STRAP
MA5,6,7: CPU CLOCK Select 1066MHz
MA4,10: DRAM CLOCK Select 366MHz
Disable JTAG
Disable IDE2
JP1
OPEN
PCB NO. DM133A
Vortex86MX EVB
VER :0.4
DMP ELECTRONICS INC.
Vortex86MX DDR-II Memory
Document Number DM133A
Rev 0.4
Thursday, November 12, 2009
Sheet 1 of 17

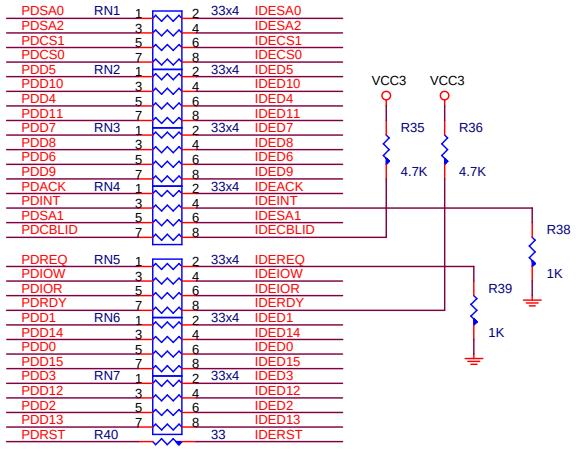


POWER LED

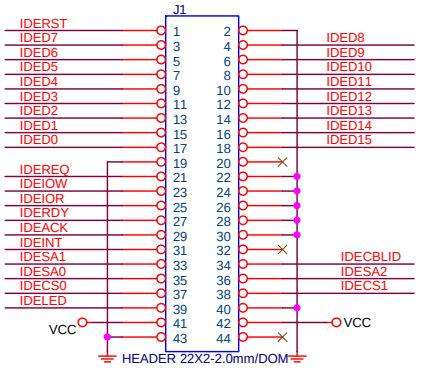


IDE LED

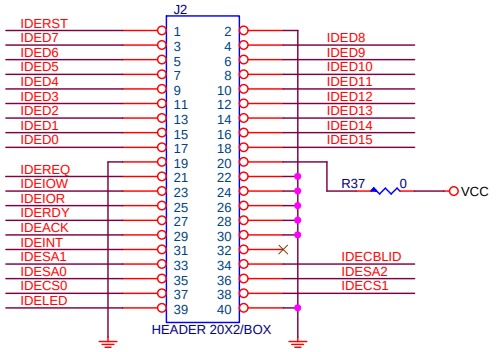




IDE 44PIN/DOM



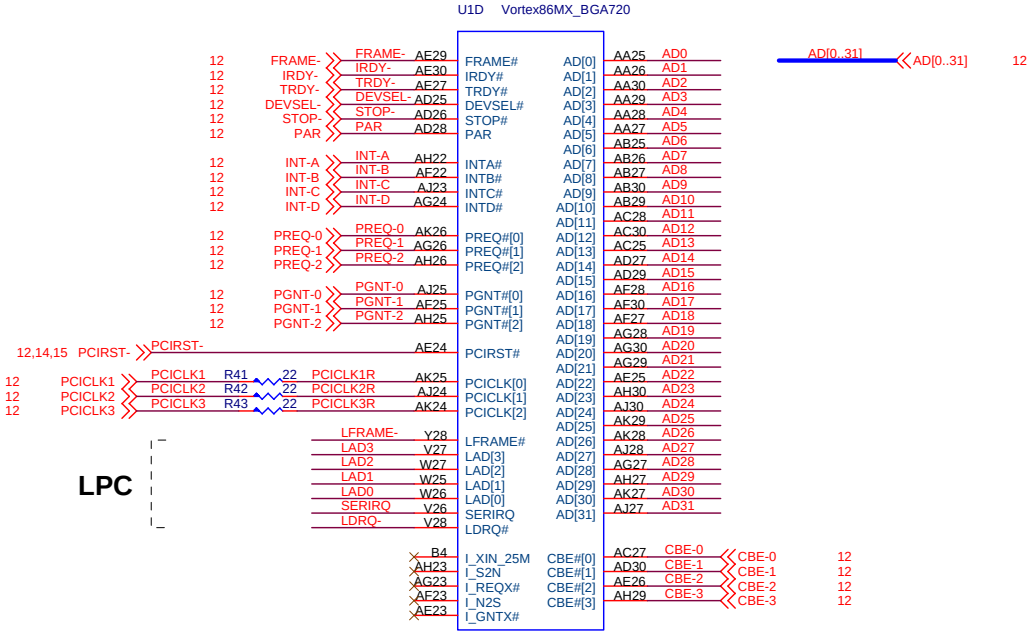
IDE 40PIN



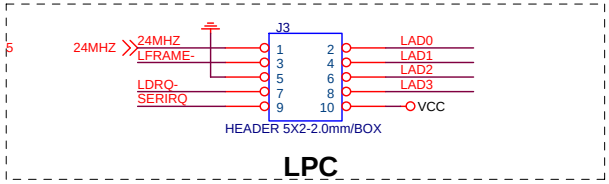
DMP ELECTRONICS INC.

Title	Vortex86MX IDE	
Size	Document Number DM133A	Rev 0.4
Date: Thursday, November 12, 2009	Sheet 2 of 17	

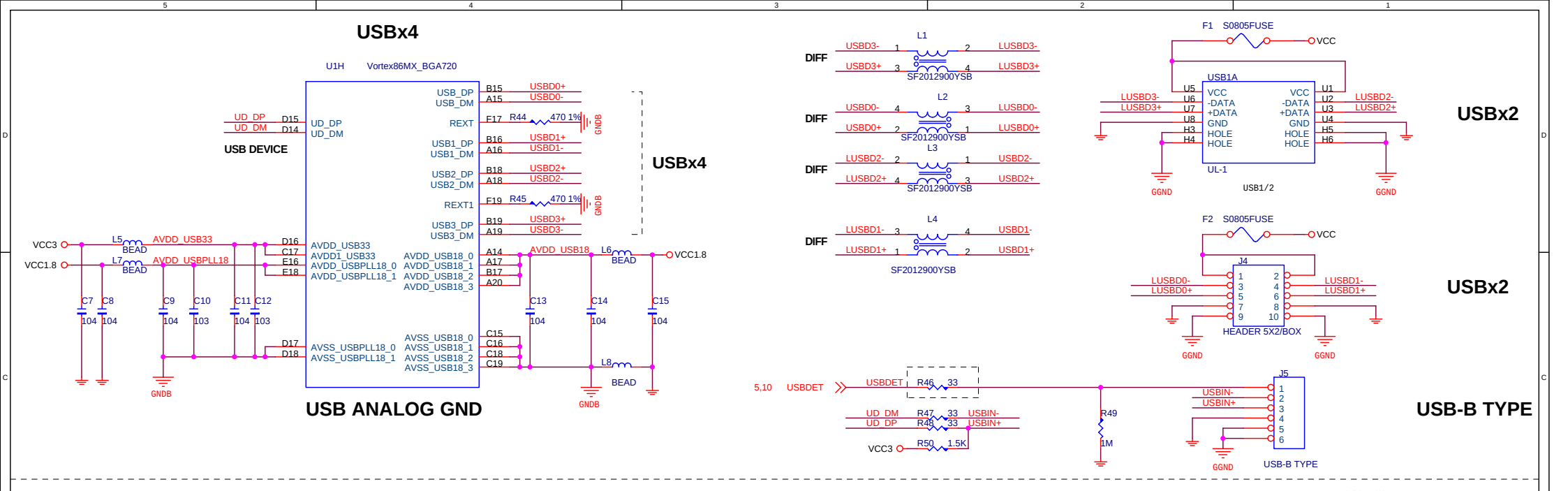
PCI BUS



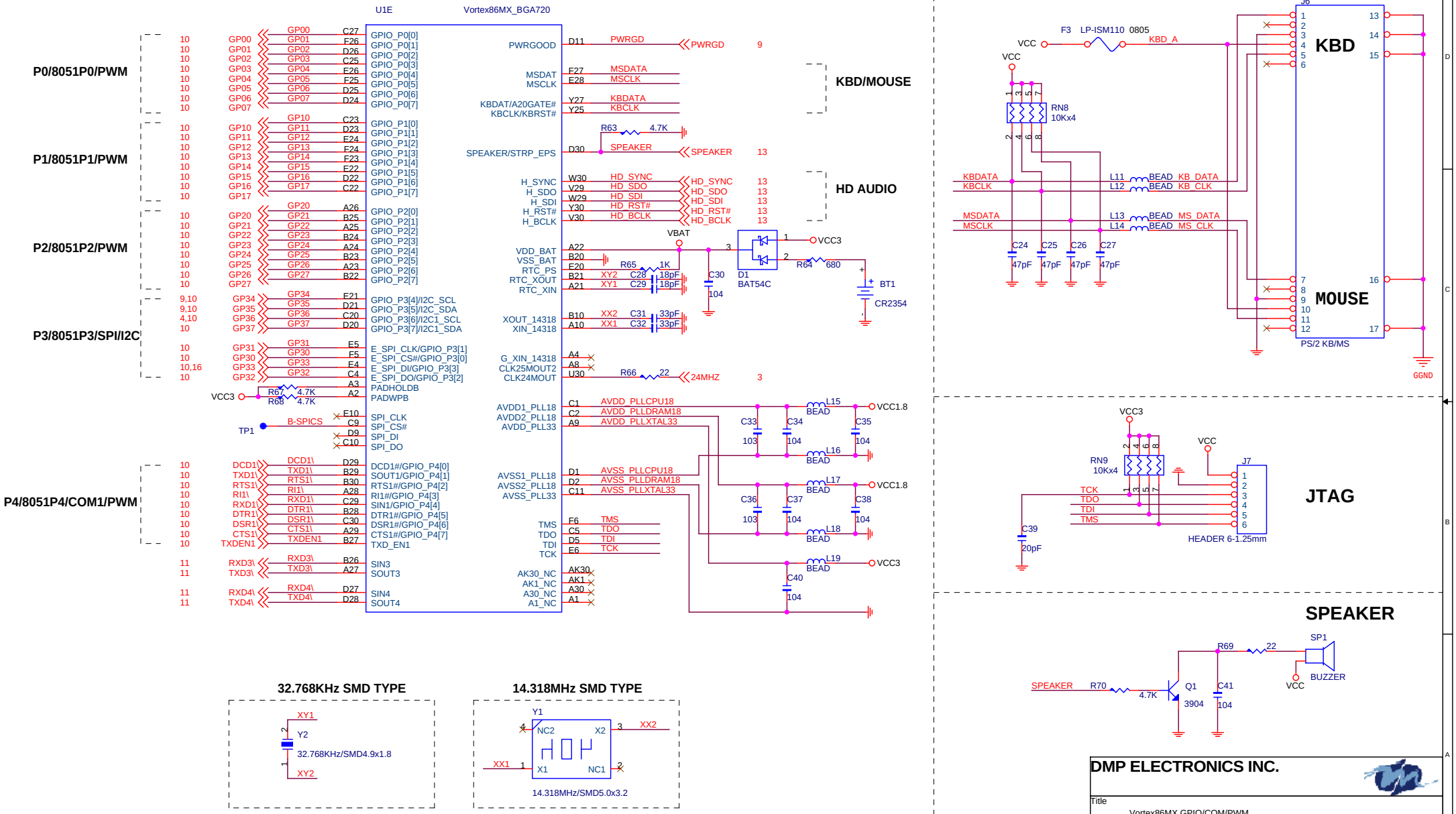
OPTION



DMP ELECTRONICS INC.			
Title: Vortex86MX PCI/ISA BUS			
Size	Document Number: DM133A		Rev: 0.4
Date: Thursday, November 12, 2009	Sheet 3 of 17		

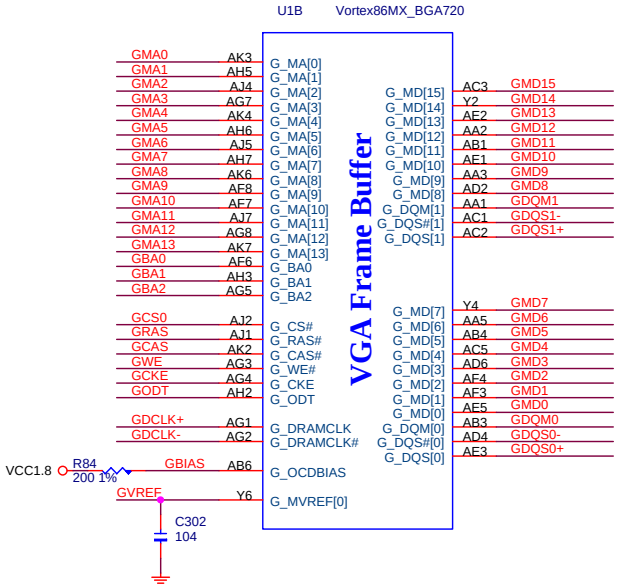


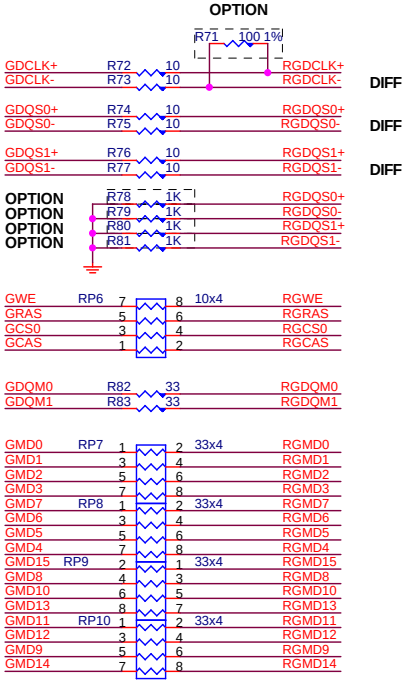
DMP ELECTRONICS INC.	
Title	Vortex86MX LAN/USB
Size	Document Number DM133A
Rev	0.4
Date:	Thursday, November 12, 2009
Sheet	4 of 17

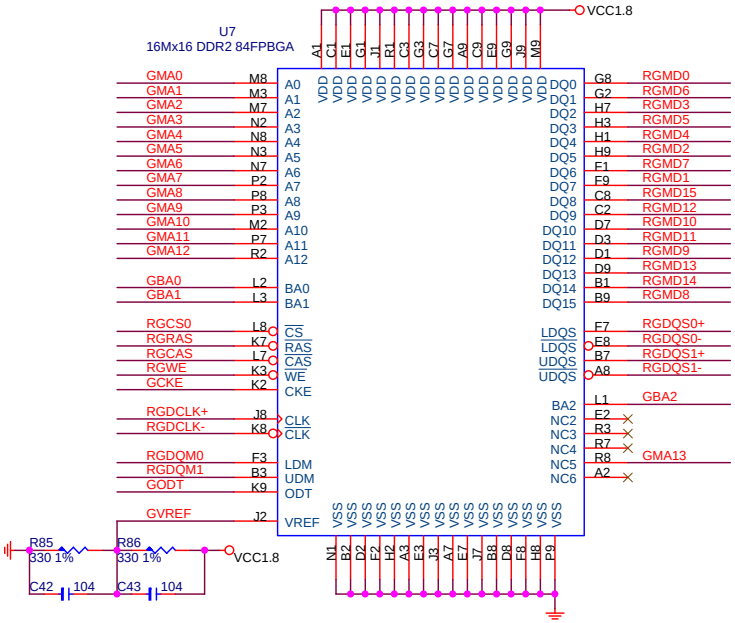


DMP ELECTRONICS INC.

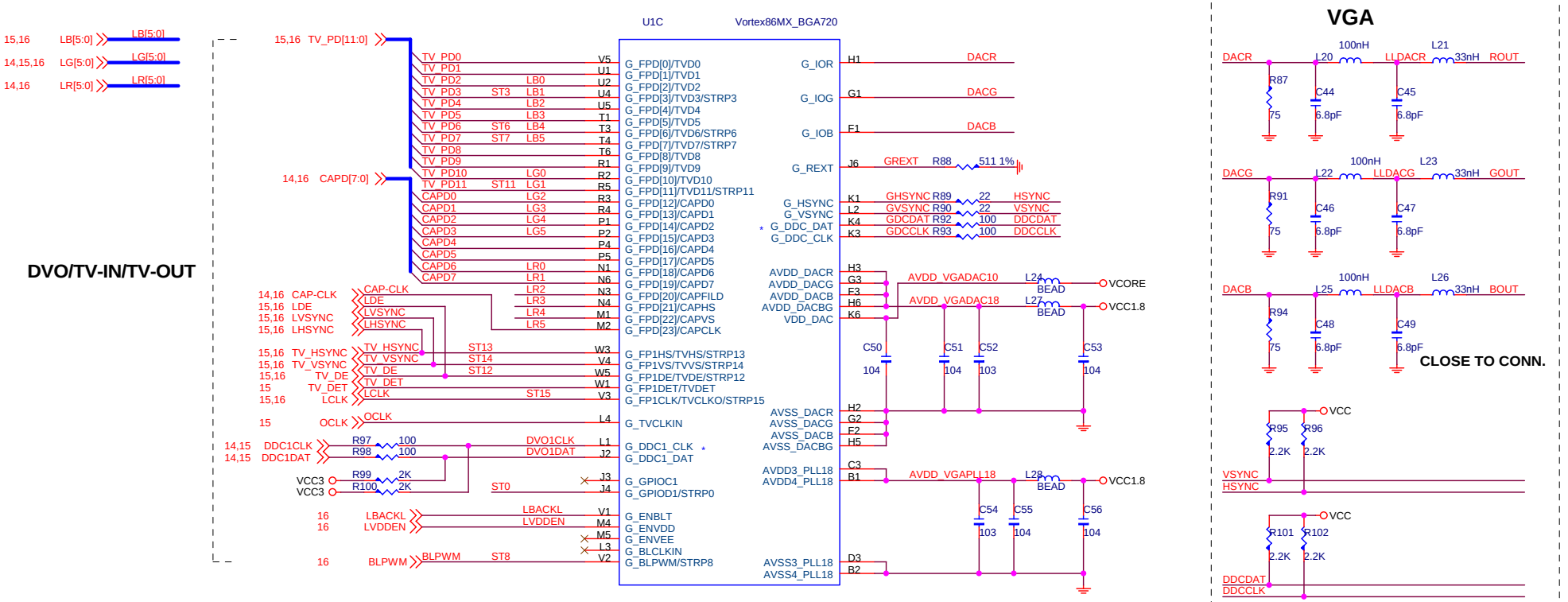
Title	Vortex86MX GPIO/COM/PWM	
Size	Document Number DM133A	Rev 0.4
Date: Thursday, November 12, 2009	Sheet 5 of 17	





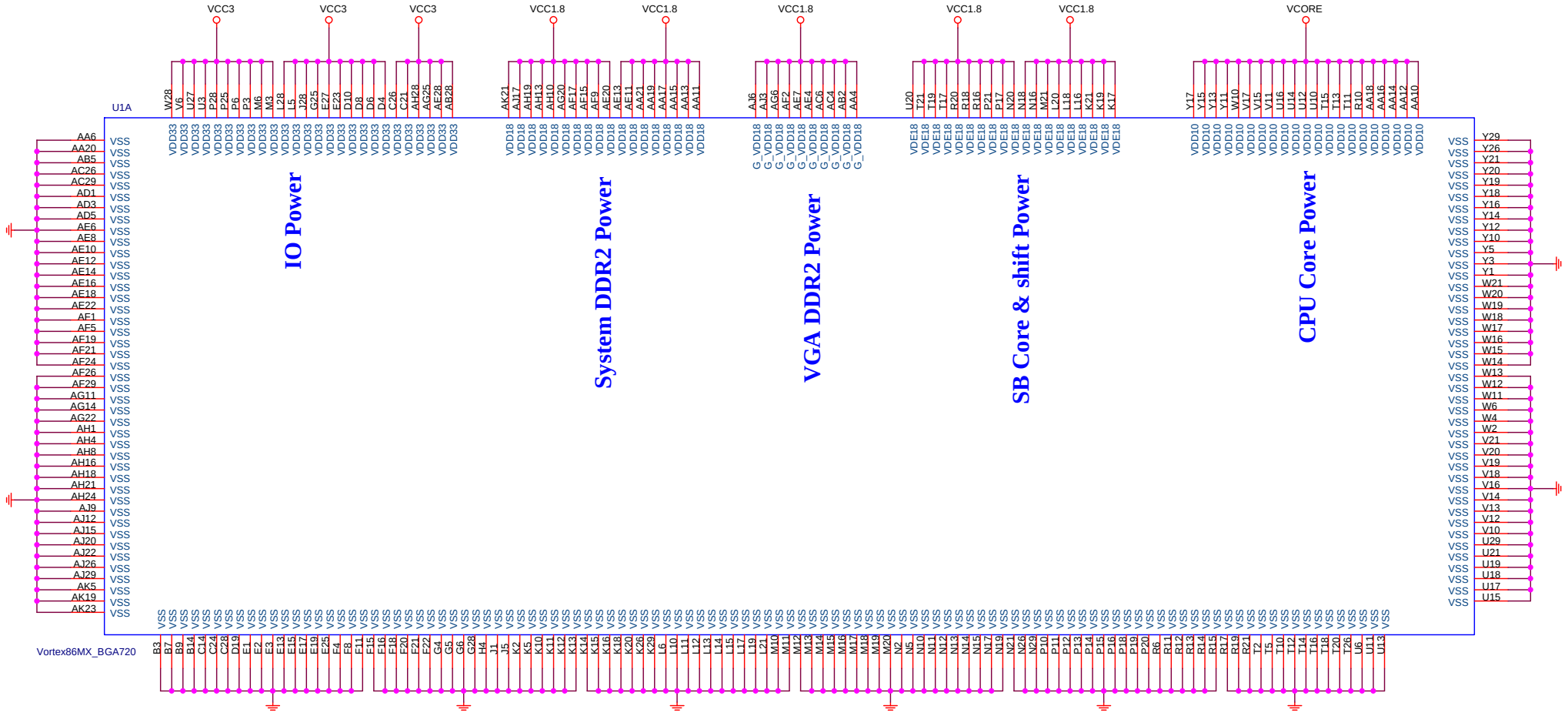


DMP ELECTRONICS INC.		
Title: Vortex86MX VGA DRAM		
Size	Document Number: DM133A	Rev: 0.4
Date: Thursday, November 12, 2009	Sheet 6 of 17	

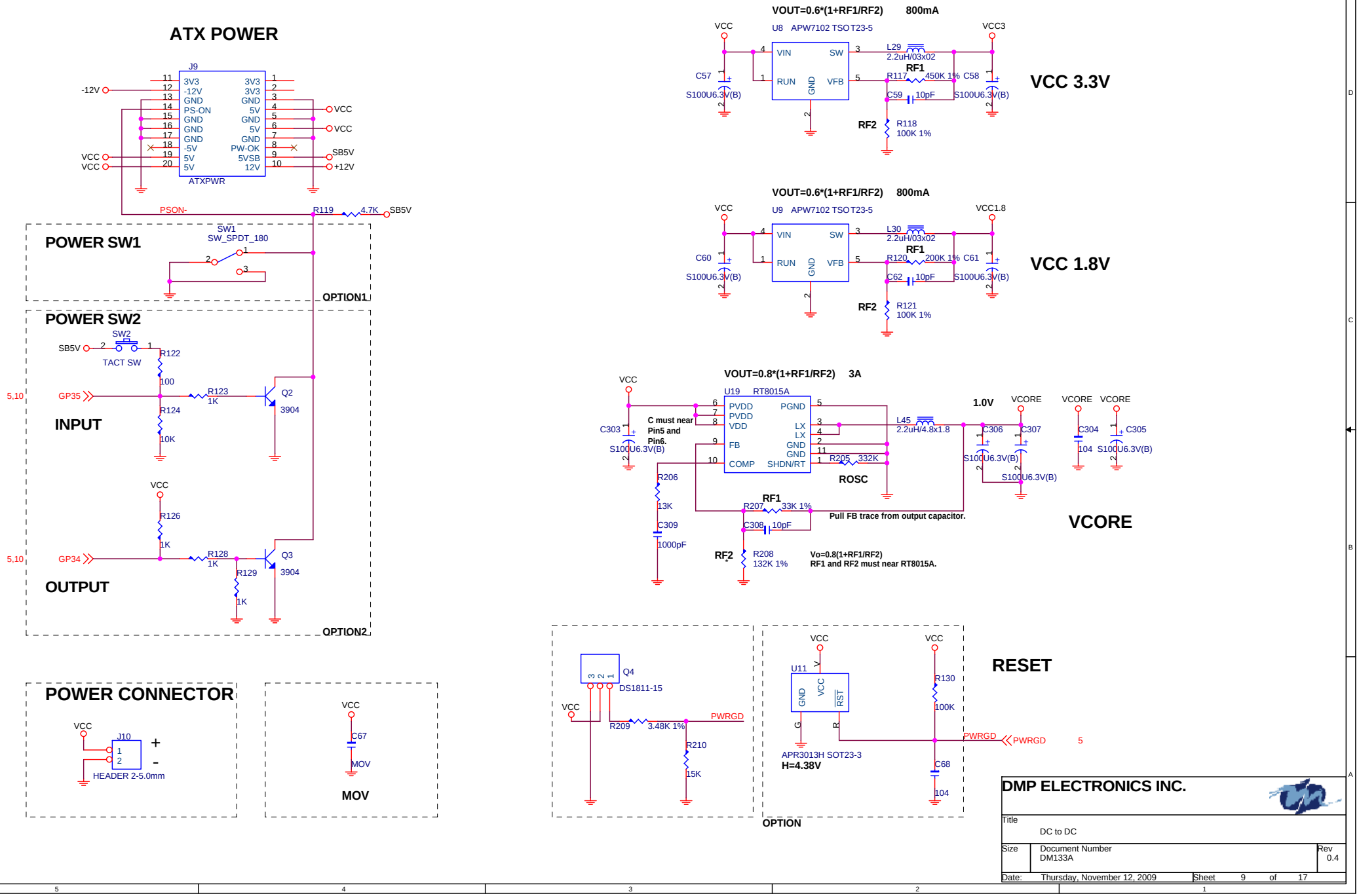


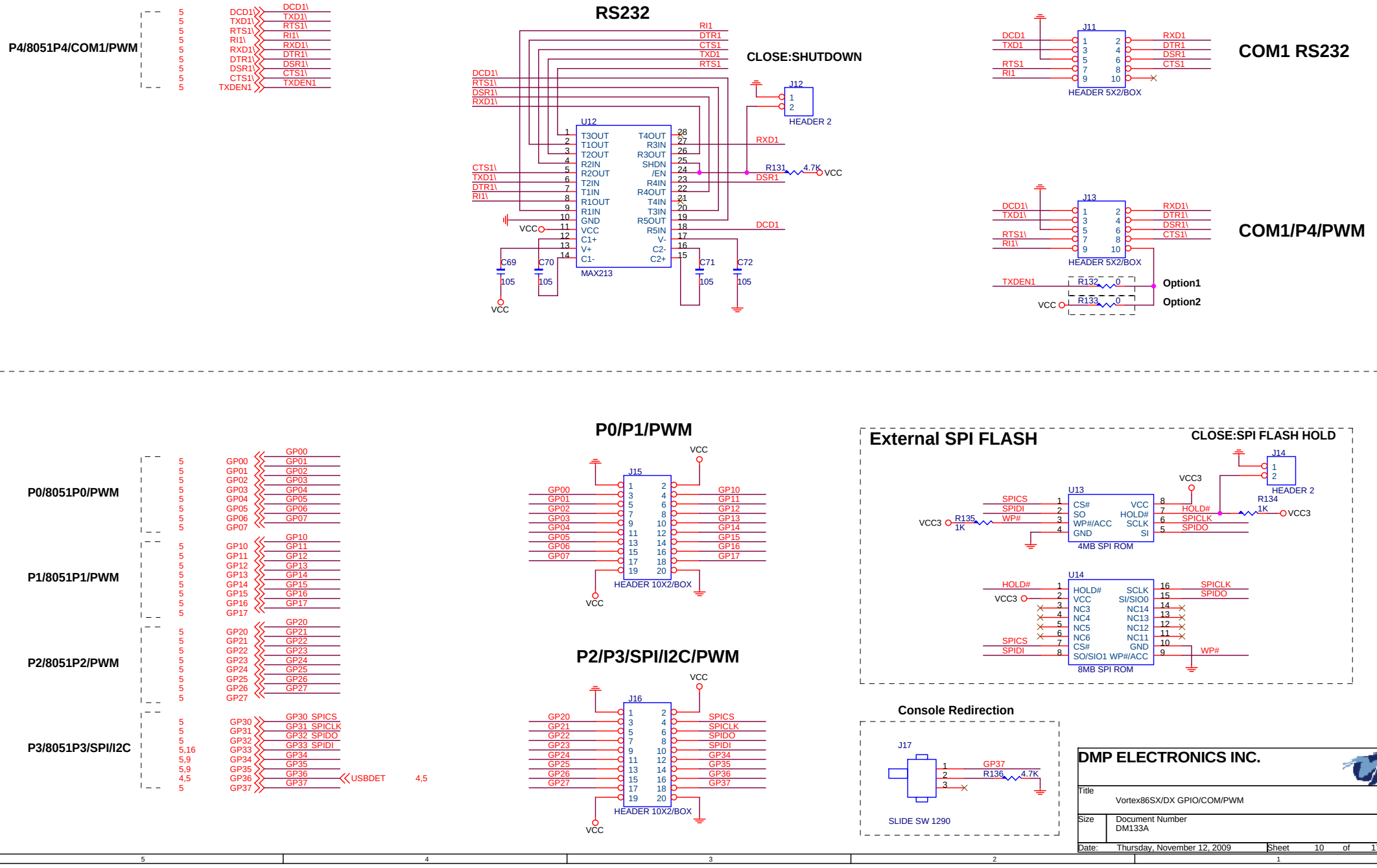
DMP ELECTRONICS INC.

Title	Vortex86MX VGA/ PLL	
Size	Document Number DM133A	Rev 0.4
Date: Thursday, November 12, 2009	Sheet 7 of 17	



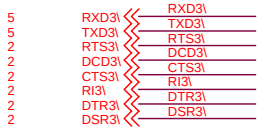
DMP ELECTRONICS INC.		
Title: Vortex86MX POWER		
Size	Document Number: DM133A	Rev: 0.4
Date: Thursday, November 12, 2009	Sheet 8 of 17	

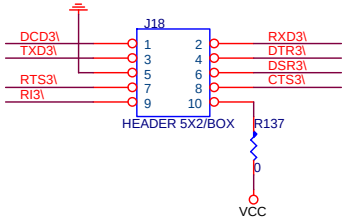

ATX POWER
J9
-12V
VCC
VCC
SB5V
+12V
ATXPWR
PSON-
R119
4.7K
SB5V
POWER SW1
SW1
SW_SPDT_180
OPTION1
POWER SW2
SW2
TACT SW
SB5V
R122
100
R123
1K
Q2
3904
5,10
GP35
INPUT
R124
10K
R126
1K
R128
1K
Q3
3904
5,10
GP34
R129
1K
OUTPUT
OPTION2
POWER CONNECTOR
J10
HEADER 2-5.0mm
C67
MOV
VOUT=0.6*(1+RF1/RF2) 800mA
U8 APW7102 TSOT23-5
VIN
SW
RUN
VFB
GND
L29
2.2uH/03x02
RF1
R117
450K 1%
C57
S100U6.3V(B)
C58
S100U6.3V(B)
C59
10pF
RF2
R118
100K 1%
VCC 3.3V
VOUT=0.6*(1+RF1/RF2) 800mA
U9 APW7102 TSOT23-5
L30
2.2uH/03x02
R120
200K 1%
C60
C61
C62
10pF
R121
100K 1%
VCC 1.8V
VOUT=0.8*(1+RF1/RF2) 3A
U19 RT8015A
PVDD
PGND
VDD
LX
FB
COMP
SHDN/RT
C must near Pin5 and Pin6.
C303
S100U6.3V(B)
L45
2.2uH/4.8x1.8
1.0V
VCORE
C306
C307
S100U6.3V(B)
C304
104
C305
S100U6.3V(B)
R205 332K
ROSC
R206
13K
C309
1000pF
RF1
R207 33K 1%
C308 10pF
Pull FB trace from output capacitor.
RF2
R208
132K 1%
Vo=0.8(1+RF1/RF2)
RF1 and RF2 must near RT8015A.
VCORE
Q4
DS1811-15
R209 3.48K 1%
PWRGD
R210
15K
U11
APR3013H SOT23-3
H=4.38V
R130
100K
C68
104
PWRGD
PWRGD 5
OPTION
RESET
DMP ELECTRONICS INC.
Title
DC to DC
Size
Document Number
DM133A
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0.4
Date: Thursday, November 12, 2009
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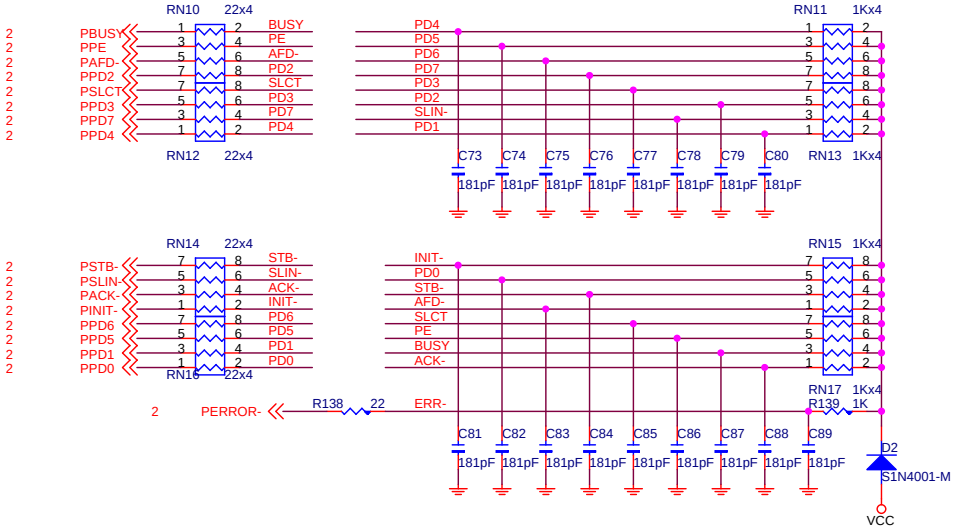

P4/8051P4/COM1/PWM
DCD1\
TXD1\
RTS1\
RI1\
RXD1\
DTR1\
DSR1\
CTS1\
TXDEN1
RS232
CLOSE:SHUTDOWN
U12
MAX213
T3OUT
T1OUT
T2OUT
R2IN
R2OUT
T2IN
T1IN
R1OUT
R1IN
GND
VCC
C1+
V+
C1-
T4OUT
R3IN
R3OUT
SHDN
/EN
R4IN
R4OUT
T4IN
T3IN
R5OUT
R5IN
V-
C2-
C2+
C69
C70
C71
C72
105
J12
HEADER 2
R131
4.7K
DSR1
COM1 RS232
J11
HEADER 5X2/BOX
COM1/P4/PWM
J13
TXDEN1
R132
R133
Option1
Option2
P0/8051P0/PWM
P1/8051P1/PWM
P2/8051P2/PWM
P3/8051P3/SPI/I2C
GP00
GP01
GP02
GP03
GP04
GP05
GP06
GP07
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP20
GP21
GP22
GP23
GP24
GP25
GP26
GP27
GP30
GP31
GP32
GP33
GP34
GP35
GP36
GP37
SPICS
SPICLK
SPIDO
SPIDI
USBDET
P0/P1/PWM
J15
HEADER 10X2/BOX
P2/P3/SPI/I2C/PWM
J16
External SPI FLASH
CLOSE:SPI FLASH HOLD
U13
4MB SPI ROM
CS#
SO
WP#/ACC
GND
VCC
HOLD#
SCLK
SI
J14
R134
R135
1K
U14
8MB SPI ROM
SI/SIO0
SO/SIO1
NC3
NC4
NC5
NC6
NC14
NC13
NC12
NC11
Console Redirection
J17
SLIDE SW 1290
R136
4.7K
DMP ELECTRONICS INC.
Title
Vortex86SX/DX GPIO/COM/PWM
Size
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0.4
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Thursday, November 12, 2009
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of
17

PRINT1/COM3/COM4

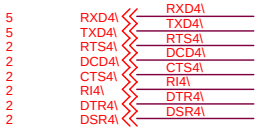
COM3 TTL

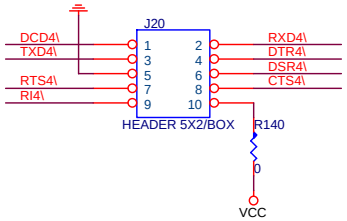


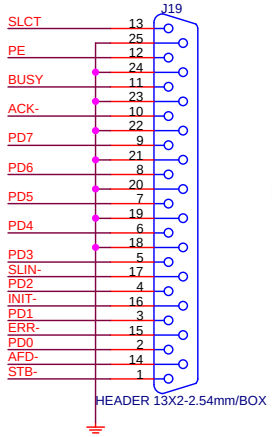




COM4 TTL



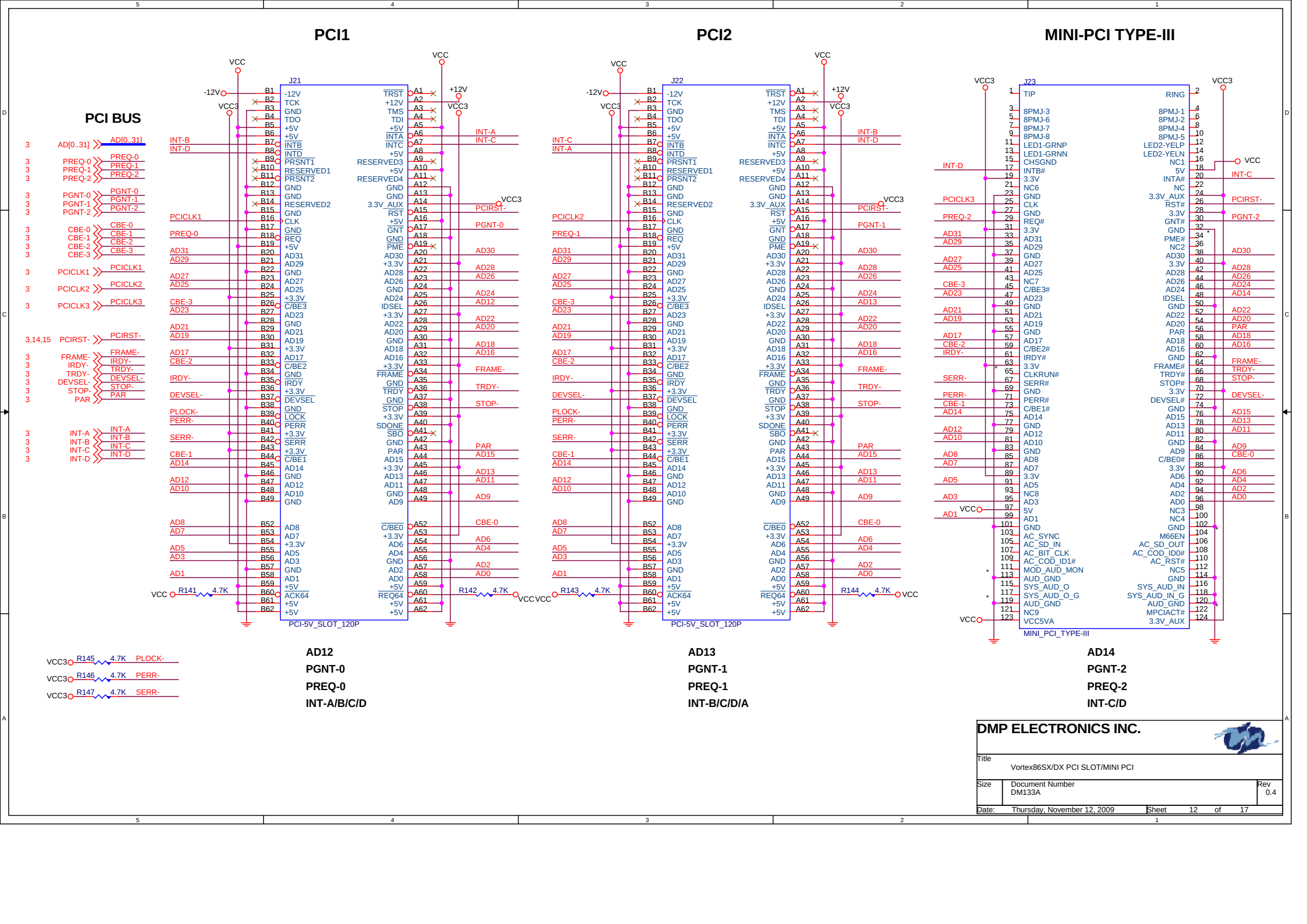




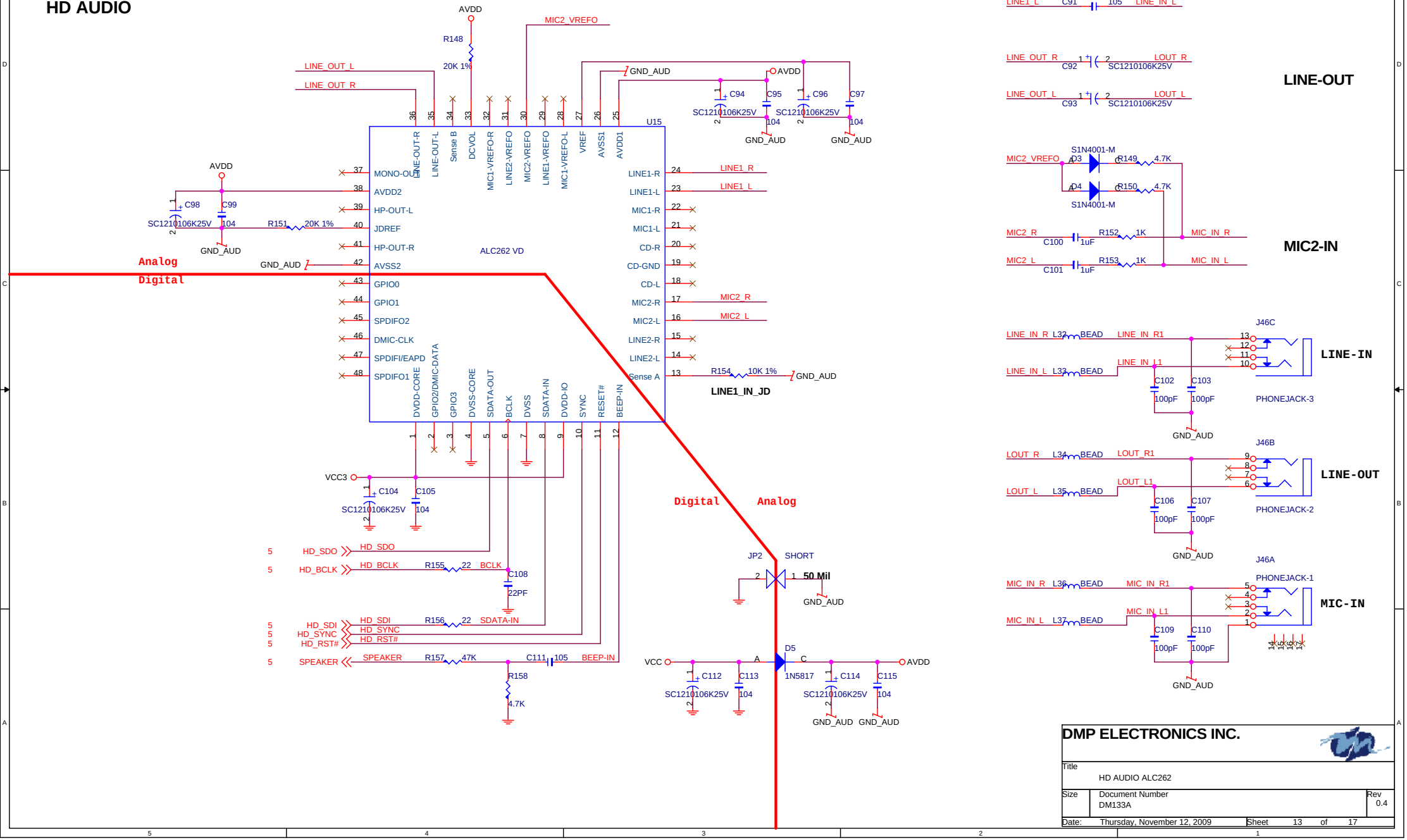
PRINT1

DMP ELECTRONICS INC.

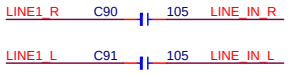
Title	Vortex86SX/DX PRN1/COM3/4	
Size	Document Number DM133A	Rev 0.4
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PCI1
PCI2
MINI-PCI TYPE-III
PCI BUS
AD[0..31]
PREQ-0
PREQ-1
PREQ-2
PGNT-0
PGNT-1
PGNT-2
CBE-0
CBE-1
CBE-2
CBE-3
PCICLK1
PCICLK2
PCICLK3
PCIRST-
FRAME-
IRDY-
TRDY-
DEVSEL-
STOP-
PAR
INT-A
INT-B
INT-C
INT-D
J21
J22
J23
PCI-5V_SLOT_120P
MINI_PCI_TYPE-III
R141 4.7K
R142 4.7K
R143 4.7K
R144 4.7K
R145 4.7K PLOCK-
R146 4.7K PERR-
R147 4.7K SERR-
AD12
PGNT-0
PREQ-0
INT-A/B/C/D
AD13
PGNT-1
PREQ-1
INT-B/C/D/A
AD14
PGNT-2
PREQ-2
INT-C/D
DMP ELECTRONICS INC.
Title
Vortex86SX/DX PCI SLOT/MINI PCI
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HD AUDIO



LINE1-IN



LINE-OUT

MIC2-IN

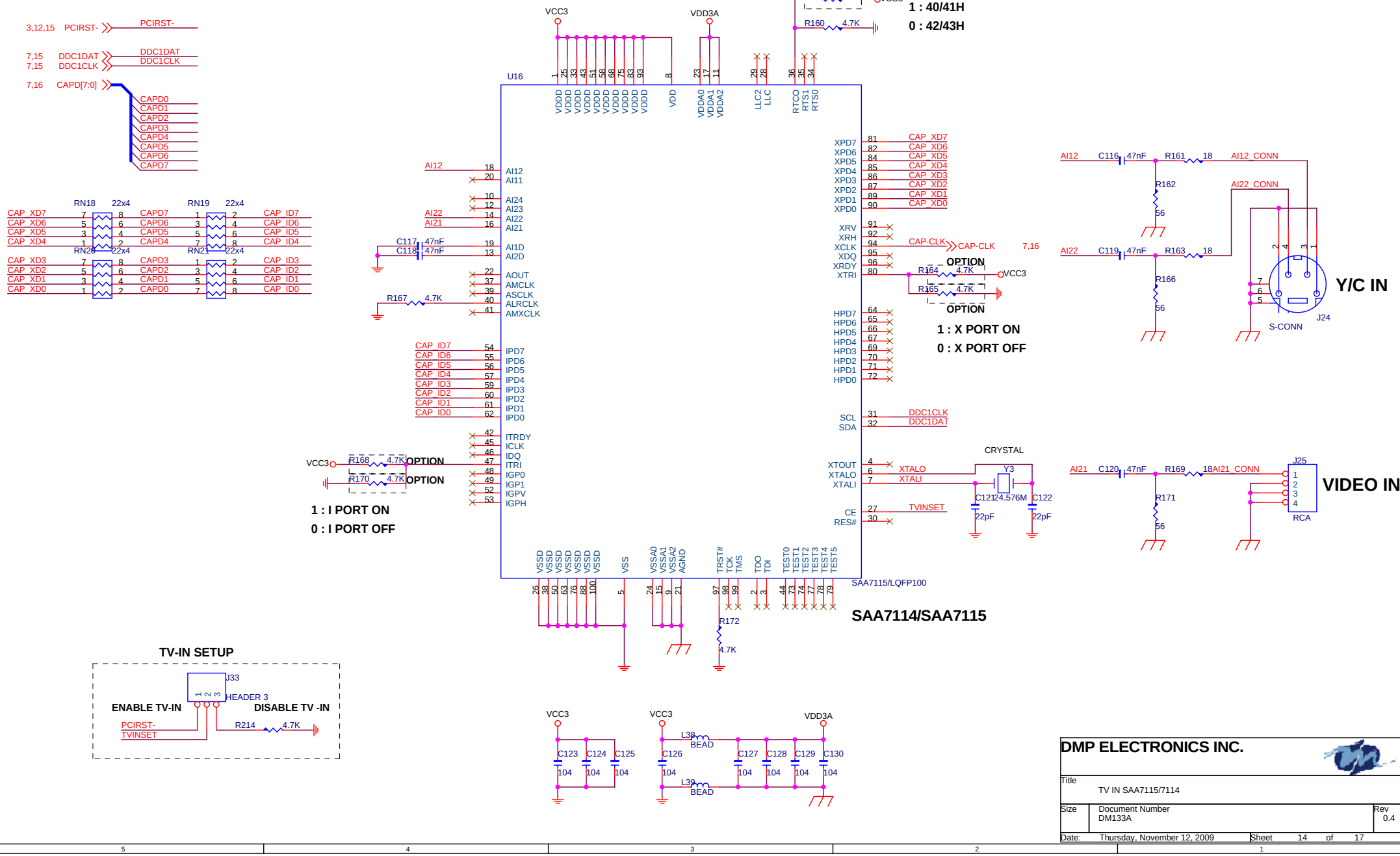
LINE-IN

LINE-OUT

MIC-IN

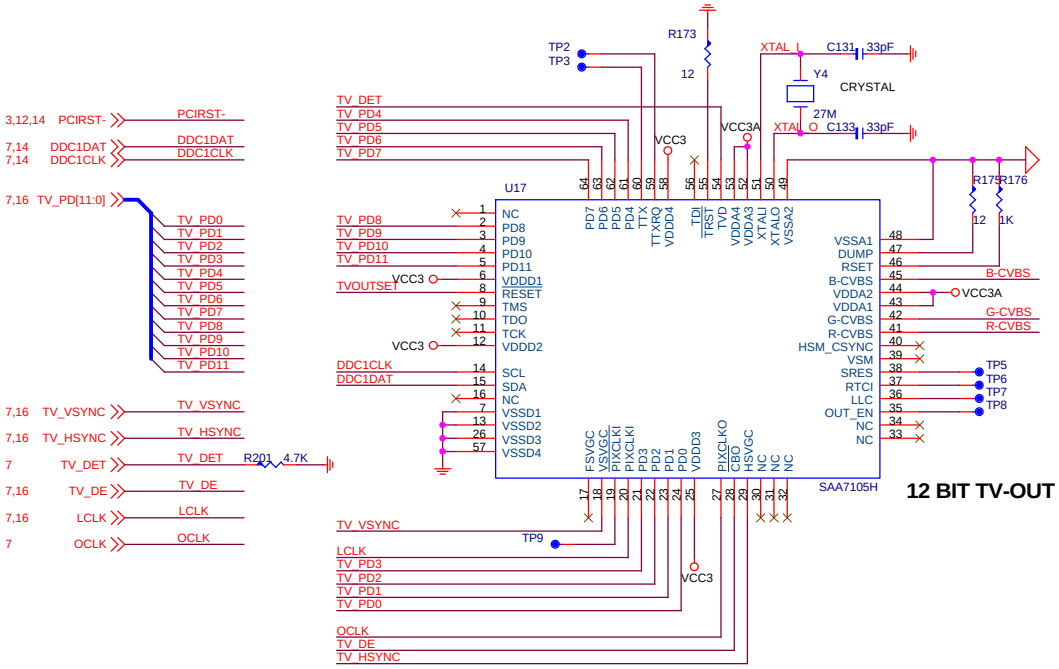
DMP ELECTRONICS INC.		
Title: HD AUDIO ALC262		
Size	Document Number: DM133A	Rev: 0.4
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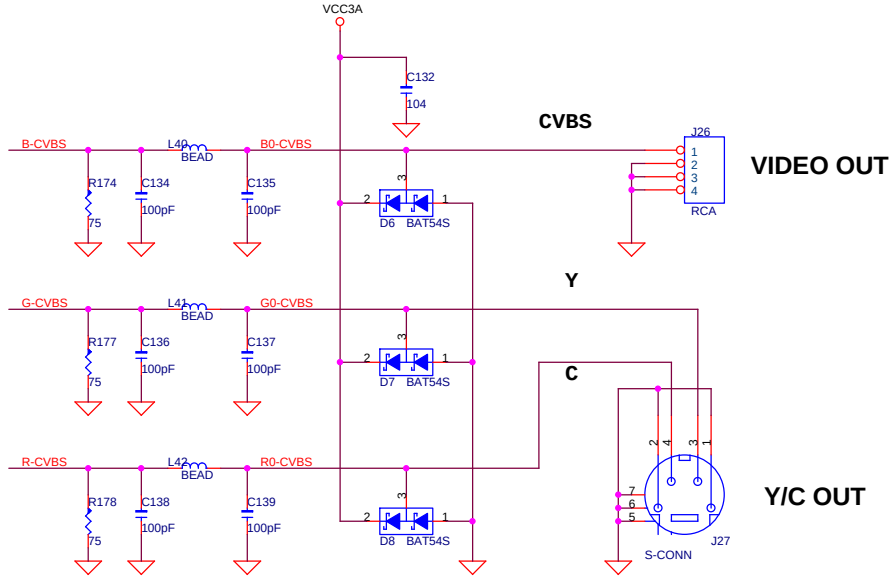
TV-IN

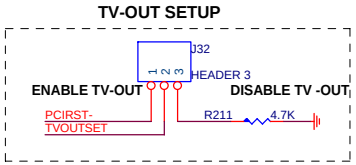


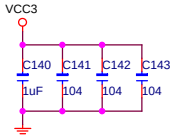
DMP ELECTRONICS INC.		
Title	TV IN SAA7115/7114	
Size	Document Number DM133A	Rev 0.4
Date: Thursday, November 12, 2009	Sheet 14 of 17	

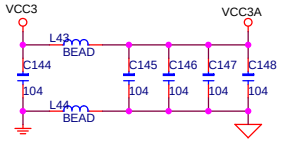
TV-OUT



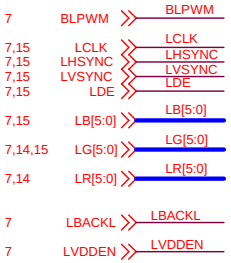


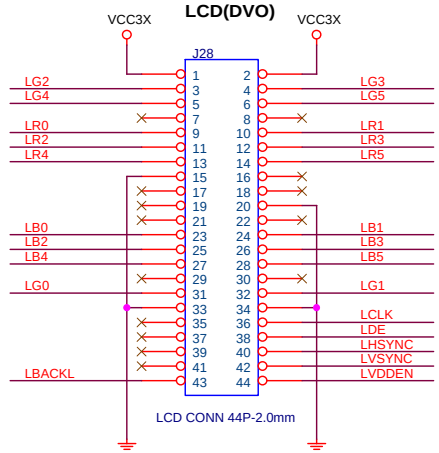


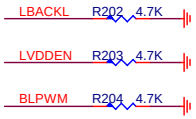




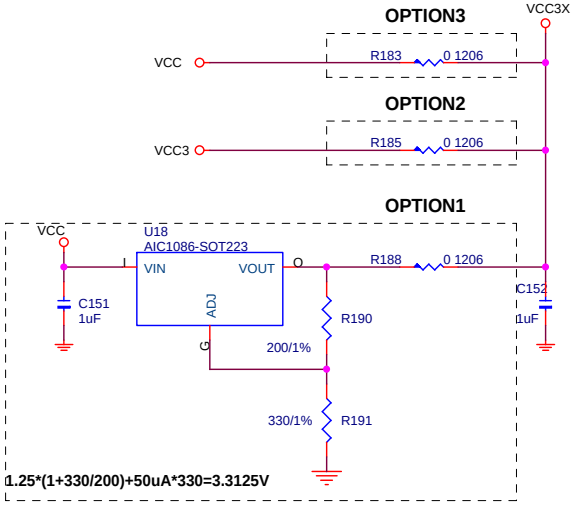
DMP ELECTRONICS INC.		
Title: TV OUT SAA7105		
Size	Document Number: DM133A	Rev: 0.4
Date: Thursday, November 12, 2009	Sheet 15 of 17	

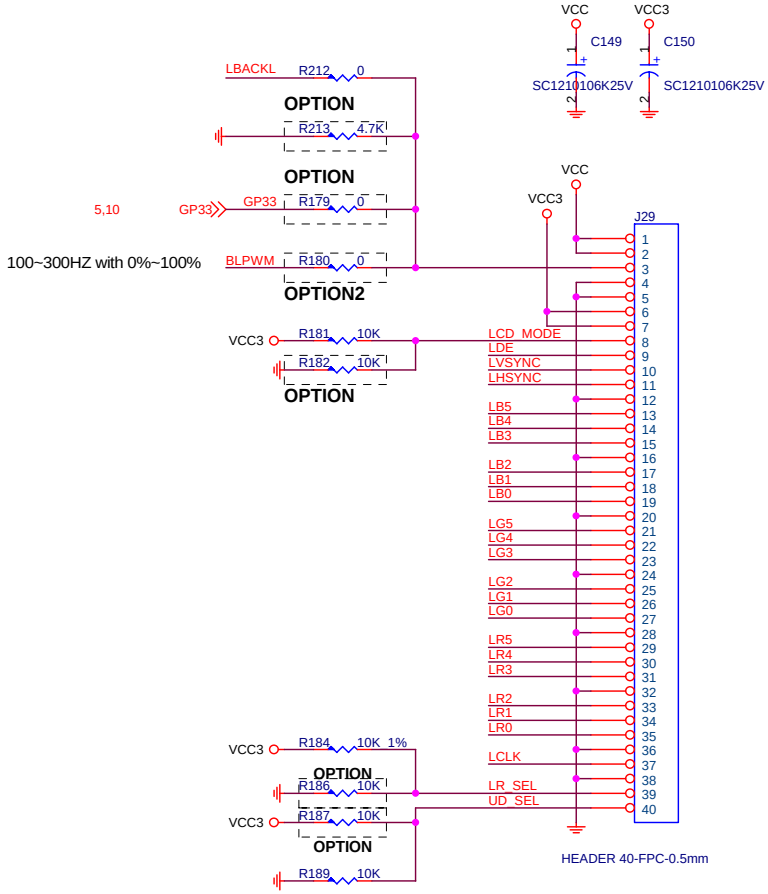






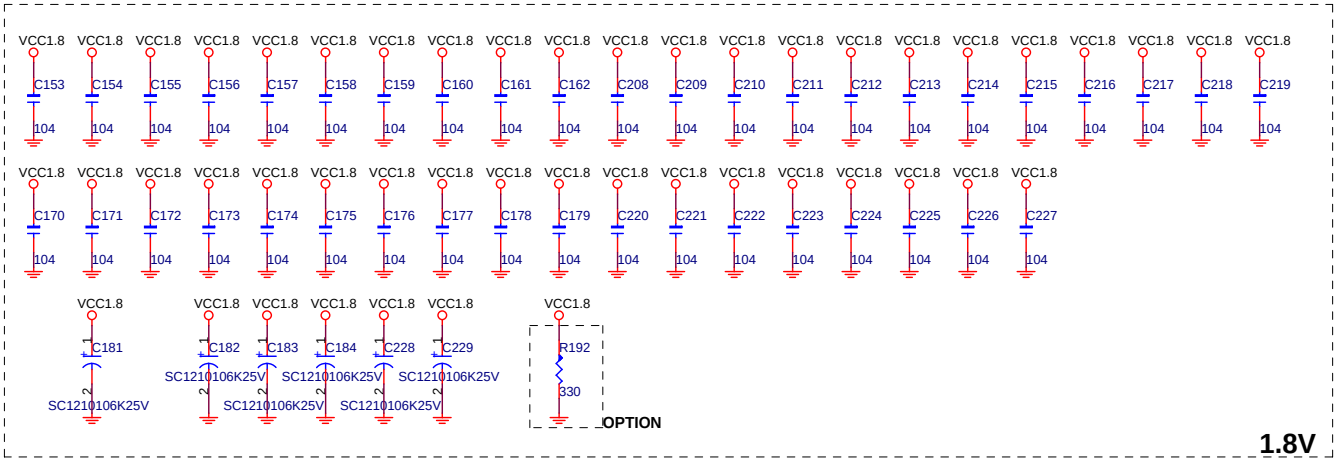
BLPWM pull-low if the backlight is darkest at BLPWM=0
BLPWM pull-high if the backlight is darkest at BLPWM=1

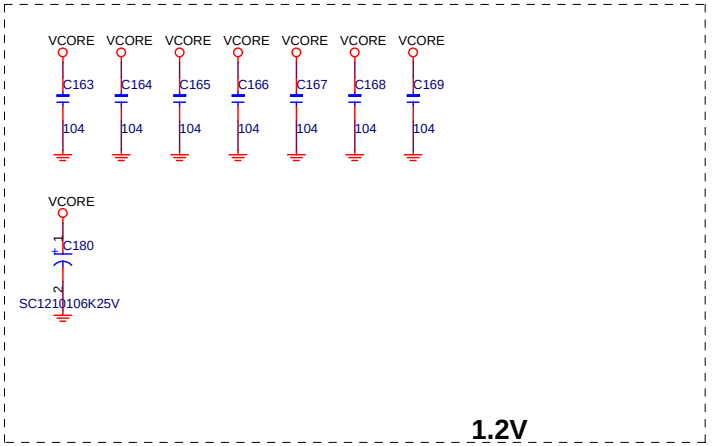


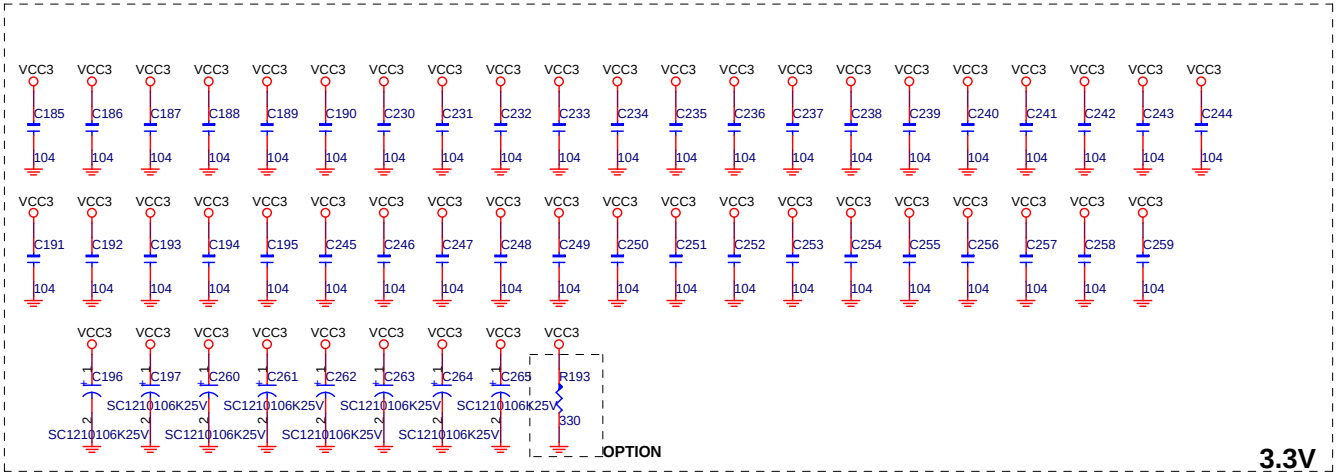


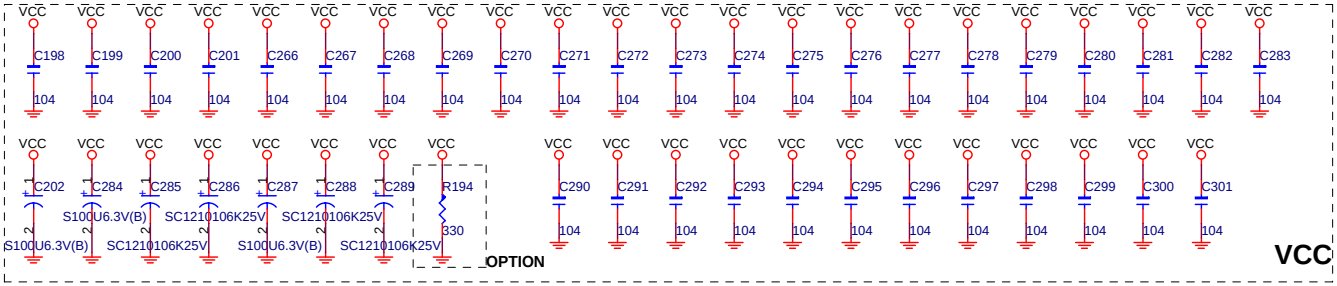
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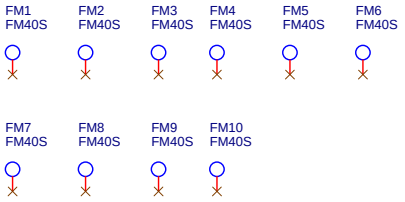
BYPASS

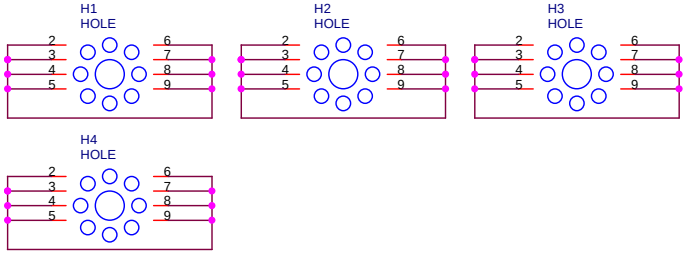












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Title: BYPASS		
Size	Document Number: DM133A	Rev: 0.4
Date: Thursday, November 12, 2009	Sheet 17 of 17	